

1.1 Introduction

Concept approach

This unit covers mechanics and materials. The unit may be taught using either a concept approach or a context approach. The concept approach begins with a study of the laws, theories and models of physics and then explores their practical applications. This section of the specification is presented in a format for teachers who wish to use the concept approach.

Context approach

This unit is presented in a different format on page 65 for teachers who wish to use a context approach. The context approach begins with the consideration of an application that draws on many different areas of physics, and then the laws, theories and models of physics that apply to this application are studied. The context approach for this unit uses three contexts for teaching this unit: sports, the production of sweets and biscuits and spare part surgery.

How Science Works

The QCA's GCE Science Criteria includes *How Science Works* (see *Appendix 6*). This should be integrated with the teaching and learning of this unit.

It is expected that students will be given opportunities to use spreadsheets and computer models to analyse and present data, and make predictions.

The word 'investigate' indicates where students should develop their practical skills for *How Science Works*, numbers 1–6 as detailed in *Appendix 6* (internal assessment may require these skills). Students should communicate the outcomes of their investigations using appropriate scientific, technical and mathematical language, conventions and symbols.

Applications of physics should be studied using a range of contemporary contexts that relate to this unit.